Research Statement

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The long-term goal of my research is to build evolvable computing infrastructure that can adapt to emerging hardware technologies and application demands. My work addresses fundamental challenges in memory and storage systems, where growing resource demands increasingly strain existing system architectures for scalability and cost efficiency.

Emerging applications, such as large language models and AI agents, are placing unprecedented demands on computing infrastructure. Their rapidly growing memory and storage requirements are making infrastructure resources increasingly scarce and expensive, while constraining system scalability. In response, the semiconductor industry is developing new hardware technologies and expanding manufacturing capacity to meet these growing demands. These advances also drive innovations in system software that can scale with evolving hardware to deliver the necessary capacity and throughput.

To address these challenges, my current research focuses on advancing memory and storage systems for cloud data centers to deliver high performance and efficiency through architectural innovations. My work has been published at leading systems conferences, including EuroSys'24, SOSP'25, FAST'26 (Best Paper Award), and OSDI'26, and has been deployed at scale in industry and adopted by the open-source community. Looking ahead, I aim to explore how AI agents will reshape both the memory and storage systems we build and the way we build them.

1. Rearchitecting Storage Systems to Unlock Hardware Capabilities

Modern storage systems face growing demands for capacity, performance, and cost efficiency. Although emerging hardware provides significant improvements in these areas, these advantages do not automatically translate into real-world benefits because of the inefficiency of existing storage architectures. My research addresses these challenges by rearchitecting storage systems across multiple aspects, from storage tiering through CPU scheduling to hardware acceleration.

CSAL: A New Storage Abstraction for High-Density SSDs [1]. High-density SSDs (e.g., QLC SSDs [2]) offer greater capacity at lower cost than previous SSDs, but face challenges in endurance and performance due to the inherent limitations of high-density flash and increased write amplification caused by coarser-grained indirection units required to reduce memory footprint. These challenges limit their deployment at scale. CSAL introduces a new storage abstraction that places high-performance SSDs between applications and high-density SSDs to reshape user writes into flash-friendly access patterns and absorb frequently updated data, achieving both high performance and endurance. CSAL enabled the deployment of high-density SSDs in large-scale cloud data centers [3] and has since been adopted by multiple production storage systems [4, 5, 6, 7, 8].

CSAL differs from traditional caching and tiering in two ways. First, it moves SSD firmware-level indexing into host software and integrates it with tiering metadata to create a unified indexing structure that enables finer-grained indexing. It further reduces memory footprint by adapting page-table and swapping techniques to the index. Second, it adopts log-structured data management across both fast and slow tiers, transforming small random writes into flash-friendly writes on high-density SSDs and managing firmware-level flash garbage collection in software.

Sandman: A New CPU Scheduler for Large-Capacity Storage Servers [9]. Recent advances in storage servers have focused on delivering higher performance and greater capacity. However, achieving high performance often comes at the cost of increased energy consumption, as modern

storage systems require a growing number of dedicated polling CPU cores for I/O processing, limiting their scalability. Sandman is a CPU scheduler designed for modern storage systems that delivers both high performance and energy efficiency. Sandman establishes CPU-side energy efficiency as a new dimension of storage sustainability in large-scale cloud environments.

Sandman makes two unique contributions compared with existing CPU schedulers. First, to rapidly scale compute resources, Sandman leverages hardware cache coherence to wake sleeping CPU cores, avoiding both software and hardware wakeup overhead. Second, to detect I/O bursts at the microsecond timescales, Sandman monitors incoming I/O requests from network queues and builds a burst-detection model based on the number of incoming I/O requests, enabling finer-grained burst detection while avoiding the inaccuracy introduced by CPU cycle-based detection.

Ristretto: An ASIC/SoC Co-Designed Stack for Cloud Storage [10]. Storage hardware performance has improved rapidly. However, modern cloud storage stacks in compute servers are increasingly constrained by available CPU resources required to sustain such performance. Ristretto introduces a new storage architecture that combines ASIC acceleration with ARM SoCs to achieve high performance while preserving flexibility. Compared with traditional CPU-based storage stacks, Ristretto delivers higher performance with substantially fewer CPU resources. Its deployment in large-scale cloud data centers demonstrates the practicality of this ASIC and SoC co-designed architecture, and the work received the Best Paper Award at FAST'26.

The key insight behind Ristretto is to rethink the hardware-software boundary by identifying which functions should be offloaded to hardware for efficiency and which should remain in software running on SoC for flexibility. Ristretto offloads compute-intensive but relatively stable functions, e.g., virtualization backends and device drivers, to ASICs, while leveraging SR-IOV to reduce virtualization overhead caused by context switches. At the same time, because cloud storage requires high flexibility for customizable features, Ristretto uses SoCs with low-power ARM cores to implement a programmable block-abstraction layer for cloud-specific functionality.

2. Designing Memory Systems Beyond Resource and Node Boundaries

Memory is a critical resource in data centers, yet traditional memory systems face two fundamental limitations: memory management focuses primarily on capacity rather than other resource dimensions and remains largely confined to individual servers. To address these limitations, my research designs new memory systems that move beyond resource and node boundaries.

RamRyder: A Software-Defined Elastic Memory System [11]. Cloud platforms offer elastic resource allocation, but memory remains inflexible: cloud providers typically allocate memory capacity in a fixed ratio to virtual CPUs and offer no mechanism for bandwidth allocation. This mismatch can lead to underutilization of both memory capacity and bandwidth as application demands diverge or change over time. RamRyder is a new memory system that enables mostly independent allocation of memory bandwidth and capacity with robust performance isolation. It establishes a new direction in software-defined, open-channel memory systems, demonstrating performance and utilization benefits using realistic cloud workload traces.

The key insight of RamRyder is to manage memory at channel granularity. Modern memory architectures provide increasing numbers of DDR channels for higher bandwidth, while CXL memory devices [12] also contain internal DDR channels that offer additional bandwidth. RamRyder shifts channel management from hardware to software, enabling direct control over channel allocation and page-to-channel mapping. To realize this design, RamRyder introduces a new OS-

level channel abstraction and page-to-channel allocation policies across memory tiers, enabling mostly independent allocation and extension of memory capacity and bandwidth.

Aspen: A New Abstraction for Efficient Memory Sharing Across Cloud VMs [13]. Cloud virtual machines (VMs) often contain redundant memory content, yet existing memory deduplication mechanisms are largely confined to individual servers, limiting memory efficiency across cloud data centers. Emerging CXL fabrics [12] enable low-latency memory sharing across distributed nodes, creating opportunities for cross-node memory deduplication. This work explores a new memory system that enables VMs on different nodes to share physical memory pages through a CXL memory pool, reducing redundant memory copies and improving memory utilization.

The key idea is to decouple VM memory management from physical memory placement in the shared pool. The system explores lightweight memory mapping and reference-based sharing mechanisms to eliminate unnecessary data duplication, together with efficient consistency management to coordinate memory access across nodes. By enabling multiple VMs to share physical memory across server boundaries, this work aims to improve memory efficiency while minimizing data movement and software overhead.

3. Future Directions: Next-Gen Memory and Storage for the Agentic Era

AI agents are becoming increasingly important and opening up a new frontier in computer systems research. They not only drive specialized system designs for agentic infrastructure but also offer new tools for building and optimizing systems. Building on my prior research, I will pursue three research directions toward next-generation memory and storage for the agentic era.

Datacenter-Scale Unified Memory for Heterogeneous Computing. Agentic AI relies on heterogeneous computing resources (e.g., CPUs, GPUs, and TPUs) for LLM reasoning and tool use to solve complex tasks. As the capabilities of LLMs and AI agents continue to grow, computing is increasingly constrained by the memory capacity of a single node or even an entire rack. These memory constraints slow down LLM inference and agent execution while leaving expensive computing resources underutilized.

To overcome this limitation, I will investigate unified memory systems that can scale across entire datacenters to support heterogeneous computing resources. First, I will explore how to efficiently interconnect memory resources by leveraging emerging scale-up (e.g., NVLink and CXL) and scale-out (e.g., Ethernet/InfiniBand RDMA) networking technologies. Second, I will investigate memory management across the memory hierarchy (e.g., HBM, DRAM, and flash) and distributed nodes, enabling transparent data movement across multiple tiers and nodes to meet the capacity, bandwidth, and latency requirements of diverse agentic workloads. Third, I will explore new protocols and algorithms that can provide scalable memory coherence across distributed nodes to enable efficient data sharing throughout the datacenter.

Rethinking Storage Architectures, Interfaces, and I/O Systems for AI Agents. Today's storage systems are largely designed for traditional cloud workloads, e.g., databases, big data analytics, and serverless computing. However, the rise of AI agents introduces new challenges for storage systems, as agents can dynamically create and terminate execution environments across cloud and edge, generate large amounts of intermediate and persistent context data, and concurrently access large shared datasets. These emerging demands not only pose new challenges to performance but also require new agent-native storage systems. To address these challenges, I will investigate new storage architectures, interfaces, and I/O systems for AI agents. First, I will

investigate new architectures that can efficiently manage large volumes of context data across distributed agents. Second, I will design new interfaces that enable secure data sharing and concurrent access among collaborating agents across cloud and edge. Third, I will develop new I/O systems that schedule I/O requests based on the characteristics of agentic workloads to achieve low latency and high resource utilization.

Harnessing AI Agents for System Engineering and Optimization. System software must continuously evolve to support emerging hardware and applications. However, developing new systems or adding features to existing ones requires substantial human effort and long development cycles. Today's AI agents have demonstrated strong capabilities in coding and research across various domains, but system software remains highly complex, where even small implementation errors can have severe consequences at the datacenter scale.

My goal is to harness AI agents to accelerate system development for emerging hardware and enable automatic optimization based on workload characteristics. To achieve this goal, I will investigate three directions. First, I will explore how to raise the abstraction level of system implementation by decoupling core system components from specialized features, allowing AI agents to operate at a higher level without compromising system stability. Second, I will investigate the use of system specifications as "prompts" to help AI agents understand system requirements and solve complex systems problems. Third, I will develop new verification and testing techniques to ensure the correctness and robustness of agent-generated system code.

Summary. Overall, my prior research in memory and storage systems for large-scale cloud infrastructure provides a strong foundation for pursuing these future directions. I aim to push the frontier of systems research while developing practical technologies that can have real-world impact. I look forward to contributing to a collaborative and diverse research environment, working with students and researchers, and helping shape the next generation of computing systems.

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